

Beyond First-Order Task Vector Superposition: Orthogonal Residuals in In-Context Learning

Anonymous ACL submission

Abstract

When in-context examples span multiple tasks, large language models (LLMs) can represent and perform them simultaneously, a phenomenon known as task superposition. A common hypothesis is that the resulting hidden representation is a convex combination of the individual task vectors. Whether this describes the hidden representations of instruction-tuned LLMs, and how faithfully it holds across depth, has not been directly examined. We find that mixed-task representations closely match their predicted convex combinations across five instruction-tuned models, three task groups, and all probed layers, establishing linear superposition as a scale-accurate first-order description. We then characterize the orthogonal residual ignored by this first-order approximation. Although its magnitude increases with depth, its resolvability above a sampling-noise floor peaks in the middle layers and depends on task separability, revealing where the linear approximation is most faithful. Finally, causal ablations show that this residual is behaviorally meaningful: removing its systematic component changes the model’s next-token distribution substantially more than matched random orthogonal perturbations as layer depth increases. Together, these results show that task superposition in instruction-tuned LLMs is well described by a linear first-order geometry whose higher-order structure emerges systematically across depth.

1 Introduction

In-context learning (ICL) enables large language models to adapt to new tasks from only a handful of demonstrations, without updating model parameters (Brown et al., 2020). Recent work has shown that in-context learning induces latent task representations, or task vectors, within transformer hid-

den states (Hendel et al., 2023; Dong et al., 2026). Building on this perspective, Xiong et al. (2025) introduced task superposition, hypothesizing that mixed-task prompts produce representations that are linear combinations of individual task vectors. While this hypothesis is supported by behavioral evidence and visualization, a mechanistic explanation for task superposition at the representational level remains an open question.

Verifying these representations matters for understanding how transformers combine multiple tasks internally and for representation steering. Existing evidence for task superposition is largely behavioral, leaving open whether the linear-combination hypothesis fully describes the hidden representations of instruction-tuned models. We make three contributions:

- We directly test the linear-combination hypothesis at the representational level across three task groups and five instruction-tuned LLMs. Mixed-task representations closely match their predicted convex combinations at every probed layer, establishing linear superposition as a scale-accurate first-order description.
- We characterize the orthogonal residual omitted by this first-order description. Although the residual grows with depth, its resolvability is governed by task separability rather than magnitude, localizing where linear superposition is most faithful.
- We causally evaluate the orthogonal residual by ablating its systematic component during the forward pass. Removing this direction perturbs the next-token distribution substantially more than norm-matched random orthogonal directions at deeper layers, showing that the residual encodes behaviorally meaningful information beyond the linear prediction.

Code and data are available at the anonymous repository https://anonymous.4open.science/r/task_vector_superposition-FAFB/ for reproducibility.

2 Related work

Task Vector Geometry in In-Context Learning

Recent work has characterized in-context learning through the lens of task vectors—latent vector representations in the transformer’s residual stream that summarize the task specified by a set of in-context demonstrations. Hendel et al. (2023) introduce the task vector $\theta(S)$, derived from in-context examples S , and show that it captures the task-relevant information necessary for in-context learning. Using a test question x , a function $f(x, \theta)$ closely reproduces the behavior of the original transformer conditioned on the demonstrations S , suggesting that task vectors provide a compact representation of the inferred task.

Dong et al. (2026) build on this perspective through the *Linear Combination Conjecture*, proposing that task vectors form compressed representations of demonstrations through linear combinations of the original examples. Extending this idea, Yan et al. (2026) study inference-time ICL in small transformers trained on synthetic latent-task distributions and show that hidden states form convex combinations of learned task vectors, with coefficients corresponding to the Bayesian posterior over latent tasks. This convex regime holds only for in-distribution tasks; they attribute out-of-distribution generalization to an additional orthogonal subspace beyond it, whereas our test questions remain strictly in-distribution, so the orthogonal residual we study arises even where their convex prediction is expected to hold exactly. We instead ask whether the same convex geometry holds empirically at the activation level in instruction-tuned models, where the mixing weights are fixed by the prompt rather than inferred.

While these works study representations induced by a single task, Xiong et al. (2025) introduce *task superposition*—the ability of LLMs to perform multiple tasks simultaneously. They establish this behavior through top- K gold-token probabilities and LDA visualizations of task vectors across model families and scales. We instead verify the linear-combination hypothesis directly at the activation level in instruction-tuned models. Behavioral superposition does not itself imply that hidden representations combine linearly, and our central object is the orthogonal residual to the predicted convex combination, tracked across depth relative to a sampling-noise floor.

Supporting this depth-dependent picture, Jiang

et al. (2025) report that ICL representations become compact and discriminative in early-to-middle layers before later layers express them to incorporate the query. This offers a mechanism for the sharpening we observe in the pure-task displacements, and hence for the floor collapse that drives resolvability in our floor-driven regime (Section 4.3).

Notions of Superposition

Superposition carries two senses in deep learning: *feature* superposition, where neurons encode multiple unrelated features through representational compression (Elhage et al., 2022), and the *task* superposition of Xiong et al. (2025)—performing multiple tasks from a single in-context prompt—which is the sense we adopt throughout. The two sit at different levels of abstraction, but both describe multiple entities sharing a common latent representation.

3 Experimental Setup

3.1 Task groups and in-context example ratios

We consider three task groups—Arithmetic addition, Entity attribution to countries, and multiple-choice questions from Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2021)—chosen so that each varies one source of variation while holding the others fixed: *question format*, *semantic relation*, and *task semantics* respectively.

Arithmetic varies the *format* of the in-context examples while preserving semantic content: two randomly generated integers `op1` and `op2` are presented as a direct sum, as a multiple-choice selection (A–D), or as a verification of a proposed sum. All three formats share the same A: answer trigger, isolating format differences from the output prompt. **Entity** varies the *semantic relation*: a country is mapped to its capital, its currency, or its language, using a manually constructed dataset; unlike the other groups, no assistant trigger is used in the user prompt. **MMLU** varies *task semantics* while holding question format constant: multiple-choice questions from the World History, Law, and Machine Learning domains share a common prompt format and the Answer: trigger, eliciting a single-letter response. Prompt templates and sampling implementation details are provided in Appendix E.

Every prompt contains 3 in-context examples and 1 test question, and we vary the tasks the examples are drawn from. For each task group we consider 10 ratios: the pure ratios (3, 0, 0), (0, 3, 0),

and $(0, 0, 3)$, where all in-context examples belong to the same task (e.g. World History for MMLU, direct addition for Arithmetic), and the mixed ratios $(2, 1, 0)$, $(1, 2, 0)$, $(0, 2, 1)$, $(0, 1, 2)$, $(2, 0, 1)$, $(1, 0, 2)$, which draw on two tasks, and $(1, 1, 1)$ taking one example from each.

3.2 Contrast vectors

The residual stream activations have been used to analyze correlations with task/function vectors. Hendel et al. (2023) demonstrated that in-context examples (which they refer to as S) are represented in a single vector $\theta(S)$, which serve as the parameters which act on the test question input x . Thus, the transformer output can be abstracted as $f(x; \theta(S))$. Injecting the extracted task vectors into the residual stream with a zero shot prompt with the same test question implicitly assumes a zero-shot baseline. Here, we make the zero-shot baseline explicit by defining a contrast vector.

Let $S = [(x_1, y_1), (x_2, y_2), (x_3, y_3)]$ be the in-context examples presented in input-output pairs. (x_i, y_i) are sampled from different tasks in a single task group according to the required task ratio r . As shown in the in-context examples in Section 3.1, the x_i questions are the user prompts and corresponding y_i answers are assistant messages. Let x be the test question. Thus, $p_r(x) := [S, x]$ be the prompt with in-context examples and $p(x)$ is the corresponding zero-shot prompt with the same test question.

Define $\mathbf{h}_l(p_r(x))$ and $\mathbf{h}_l(p(x))$ as the the residual stream state at the assistant turn boundary token in layer l with prompts $p_r(x)$ and $p(x)$ respectively. Thus, we define the contrast vector $\mathbf{c}_r^l(x)$ as

$$\mathbf{c}_r^l(x) := \mathbf{h}_l(p_r(x)) - \mathbf{h}_l(p(x))$$

For each ratio r , we sample n test questions and ICL examples to generate n contrast vectors at a layer l . In the following sections, we will refer to the i^{th} test question as x_i . For each model, we collect contrast vectors at increasing layer depths proportional to the number of layers in the model.

3.3 Representational Geometry Metrics

We quantify the extent to which contrast vectors (and by extension, task vectors) superpose with a metric based on a scalar projection. For each ratio r , the centroid of the contrast vectors at layer l is

$$\mu_r^l := \frac{1}{n} \sum_{i=1}^n \mathbf{c}_r^l(x_i)$$

The linearity hypothesis posits that if k ICL examples are drawn for a single prompt from multiple tasks in a ratio r , the resulting contrast vector at layer l is a sum of the pure task centroids, weighted by the fraction of each task in the ICL examples. Formally, for a ratio $(r_1, r_2, \dots, r_{|P|})$, the predicted centroid of the contrast vectors for ratio r and layer l is

$$\hat{\mu}_r^l := \sum_{p \in P} \frac{r_p}{k} \mu_p^l$$

where r_p denotes the component of ratio r corresponding to pure ratio $p \in P$. Let R be the set of all 10 ratios and P the set of pure ratios $(3, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 3)$ (Section 3.1), whose in-context examples are sampled from a single task. The tri-center of contrast vectors at layer l is

$$\mu^l := \frac{1}{|P|} \sum_{r \in R} \mu_r^l [[r \in P]]$$

From μ_r^l , $\hat{\mu}_r^l$, and μ^l we define the displacement vectors $\mathbf{d}_r^l := \mu_r^l - \mu^l$ and $\hat{\mathbf{d}}_r^l := \hat{\mu}_r^l - \mu^l$; Figure 1 shows their relationships in 2D. From $\mathbf{d}$ and $\hat{\mathbf{d}}$, we can now define κ :

$$\kappa_r^l := \frac{\mathbf{d}_r^l \cdot \hat{\mathbf{d}}_r^l}{\|\hat{\mathbf{d}}_r^l\|^2}$$

κ is the factor by which $\hat{\mathbf{d}}_r^l$ must be scaled to match the projection of $\mathbf{d}_r^l$ onto it. So $\kappa = 1$ indicates that the convex prediction accounts exactly for the component of $\mathbf{d}_r^l$ along $\hat{\mathbf{d}}_r^l$ —a dominant linear term rather than collinearity, since any component of $\mathbf{d}_r^l$ orthogonal to $\hat{\mathbf{d}}_r^l$ leaves κ unchanged—while $\kappa = 0$ indicates complete orthogonality.

Note that for $r = (1, 1, 1)$, $\hat{\mu}_{(1,1,1)}^l = \frac{1}{3}\mu_{(3,0,0)}^l + \frac{1}{3}\mu_{(0,3,0)}^l + \frac{1}{3}\mu_{(0,0,3)}^l = \mu^l$: the predicted centroid and the tri-center coincide, so $\hat{\mathbf{d}}_{1-1-1}^l = \mathbf{0}$ and κ_{1-1-1}^l is undefined. We therefore do not compute κ for the 1-1-1 ratio.

Because it is a scalar projection, κ does not take into consideration the variations orthogonal to $\hat{\mathbf{d}}_r^l$. It can thus establish whether linear superposition captures the dominant representational structure, but not whether it completely characterizes the geometry; we use it as a first-order diagnostic before examining the orthogonal residual directly.

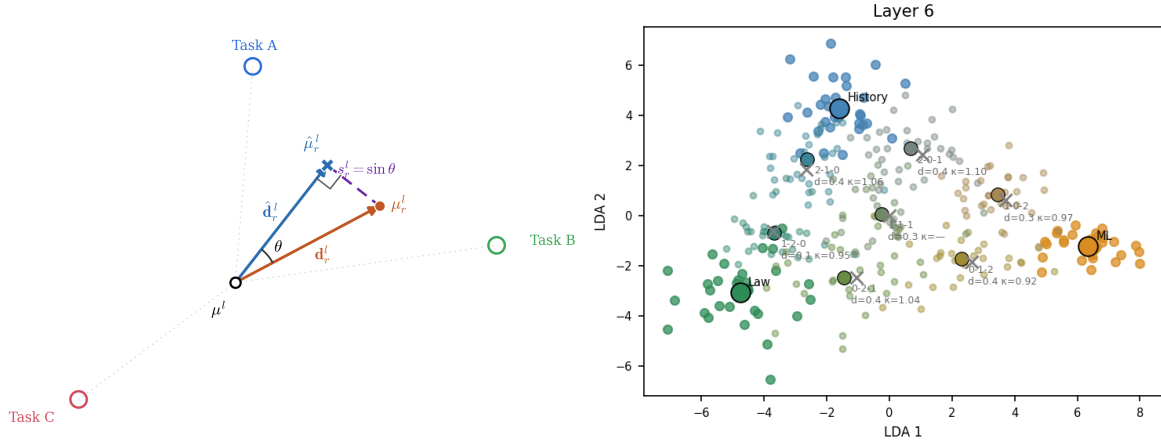


Figure 1: Geometry of the linear superposition prediction. (a) Relationship between the tri-center, pure-task centroids, and the observed versus predicted mixed-ratio centroids and contrast vectors. (b) An LDA projection of contrast vectors for the MMLU task group in Llama 3.2 3B (layer 6), showing predicted versus actual centroids for a mixed ratio. **Colors:** Blue = History, Orange = ML, Green = Law. **Markers:** $\times$ predicted centroid, $\bullet$ actual centroid for a ratio.

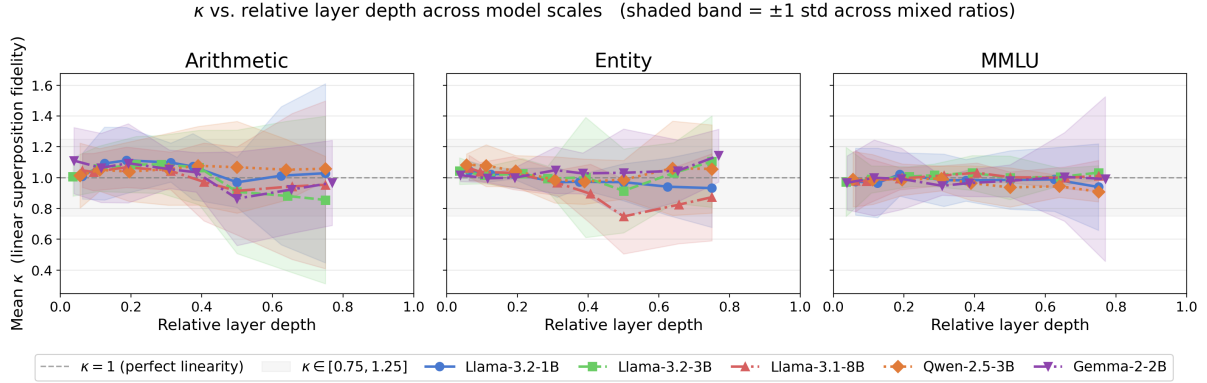


Figure 2: Cross-model κ versus relative layer depth for the Arithmetic, Entity, and MMLU task groups (panels share a y -axis). Each line is one model’s mean κ across the $K = 6$ mixed ratios; the shaded band of the same color is ± 1 standard deviation across those ratios. The dashed line marks $\kappa = 1$ (exact linear superposition) and the grey zone $\kappa \in [0.75, 1.25]$.

4 The Geometry of Mixed-Task Representations Across Depth

4.1 Geometry of mixed-task representations

In this section, we empirically analyze contrast vectors induced by varying in-context example ratios across the Arithmetic, Entity, and MMLU task groups. We evaluate this behavior across five instruction-tuned language models spanning multiple families and scales: Llama 3.2 1B, Llama 3.2 3B, Llama 3.1 8B (Grattafiori et al., 2024), Gemma 2 2B (Gemma Team, 2024), and Qwen 2.5 3B (Yang et al., 2024). Following Xiong et al. (2025), we visualize the resulting contrast vectors using an LDA projection fit on the pure-task centroids, extending their analysis of task superposition from pre-trained models to practically deployed instruction-tuned LLMs, and Figure 1 con-

tains an example of the same. Implementation details for prompt construction, relative layer selection, contrast vector extraction, and κ computation are provided in Appendix A; representative layer-wise contrast-vector visualizations are shown in Appendix F, with the complete set for every model and task group available in the accompanying code repository.

To confirm that $\kappa \approx 1$ reflects a genuine first-order linear relationship rather than sampling variability, we evaluate it using two complementary statistical procedures. First, a permutation test compares the observed κ against a null distribution obtained by randomly permuting the correspondence between mixed-task centroids and their predicted convex combinations, testing whether the observed projection fidelity exceeds chance. Second, we

compute 95% bootstrap confidence intervals over prompt samples to quantify the uncertainty in the estimated mean κ . Across all models, task groups, and sampled layers, the observed values remain centered near $\kappa = 1$, with confidence intervals consistently overlapping unity and lying well above the permutation null, establishing that a convex combination provides an accurate first-order description of mixed-task representations throughout the network.

Additionally, Figure 2 plots the mean κ as a function of relative layer depth for every model and task group. While the mean remains close to unity across depth, the variability across mixed-task ratios increases toward later layers (standard deviation increasing by 0.25 for Arithmetic, 0.19 for Entity, and 0.024 for MMLU between the first and last sampled layers), suggesting that although the dominant linear projection remains stable, a single scalar projection becomes progressively less descriptive of the full representational geometry.

4.2 The orthogonal residual and its noise floor

κ pins the component of the observed displacement *along* the predicted convex direction, but is blind to any *orthogonal* component: a centroid could project onto the prediction with unit coefficient while still lying off the predicted point. We therefore measure the orthogonal residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$ directly and calibrate it against a split-half sampling-noise floor—the angle between two independent half-sample estimates of the same displacement, i.e. the best agreement the data itself can resolve (Appendix C). Formally, for ratio r at layer l we define the orthogonal residual s_r^l , the split-half noise floor ϕ_r^l , and the size-matched residual $\tilde{s}_r^l$ as

$$s_r^l := \sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l) = \sqrt{1 - \frac{(\mathbf{d}_r^l \cdot \hat{\mathbf{d}}_r^l)^2}{\|\mathbf{d}_r^l\|^2 \|\hat{\mathbf{d}}_r^l\|^2}}, \quad (1)$$

$$\phi_r^l := \frac{1}{B} \sum_{b=1}^B \sin \theta(\mathbf{d}^{(1,b)}, \mathbf{d}^{(2,b)}), \quad (2)$$

$$\tilde{s}_r^l := \frac{1}{2B} \sum_{b=1}^B \sum_{j \in \{1,2\}} \sin \theta(\mathbf{d}^{(j,b)}, \hat{\mathbf{d}}_r^l) \quad (3)$$

where $\mathbf{d}^{(1,b)}, \mathbf{d}^{(2,b)}$ are the displacements formed from the two halves of a random equipartition b of ratio r 's n samples ($B = 500$), with the pure-task basis held at its full-sample value so only the

sampling noise in μ_r varies. We report the residual *magnitude* with the full-sample s_r^l and test *resolvability* with the size-matched $\tilde{s}_r^l$: layer l counts as *within floor* iff $\tilde{s}_r^l \leq \phi_r^l$. The full-sample s_r^l is the minimum-variance estimate, so it tracks the magnitude across depth without being flattened by sampling noise, while the size-matched $\tilde{s}_r^l$ shares the floor's $n/2$ noise for a fair resolvability test. Table 1 and Figure 3a report this across all five models and all three task groups.

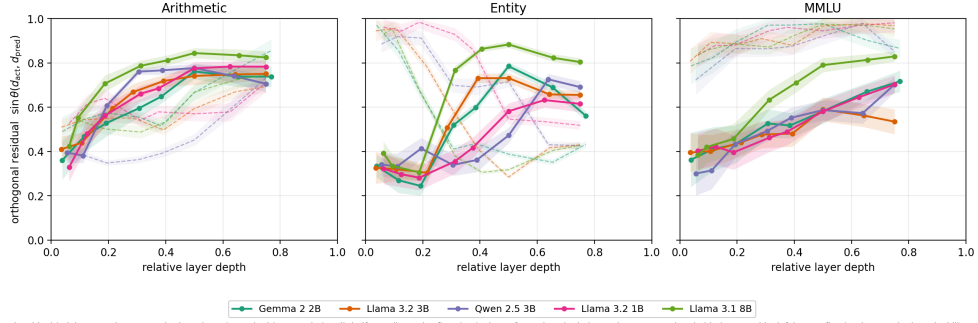
Two findings emerge. First, the orthogonal residual grows with depth in every task group, from $\sin \theta \approx 0.30$ – 0.42 at the shallowest swept layer to ≈ 0.53 – 0.83 at the deepest (Appendix C). Averaged across models, the increase is monotonic for MMLU and rises before leveling off in the Arithmetic and Entity groups. Linear superposition is therefore geometrically tightest in the early layers, where the residual lies within the sampling-noise floor in all but two of the fifteen model–task combinations.

Second, whether this growing residual becomes statistically resolvable depends on task separability rather than depth alone. For Arithmetic and Entity, increasing separation between the pure-task displacements shrinks the sampling-noise floor until the residual clears it. By contrast, MMLU's common multiple-choice format keeps its task vectors weakly separated and the floor correspondingly high, so the residual remains near the floor throughout, exceeding it only marginally at the deepest swept layers in Gemma-2B and Qwen-3B.

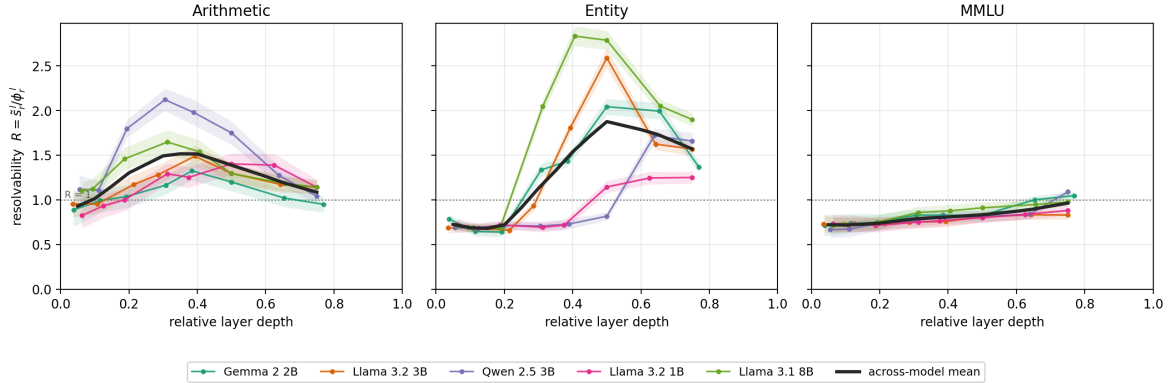
We stress that this third finding concerns resolving power, not a breakdown of linearity: MMLU's residual follows the same shallow-to-deep growth as the other groups but sits beneath a higher floor at nearly every depth, and κ remains ≈ 1 in every case. The dominant structure is the convex combination at all depths, with an additional, secondary orthogonal component becoming resolvable only where the geometry is separable enough to expose it.

4.3 Resolvability of the orthogonal residual across depth

We plot the resolvability ratio $R_r^l := \tilde{s}_r^l / \phi_r^l$ —the size-matched orthogonal residual over the split-half sampling-noise floor—as a function of relative layer depth, aggregated across models within each task group (Figure 3b). Here $R = 1$ is the noise floor: $R > 1$ marks a statistically resolvable orthogonal component, whereas $R \leq 1$ means the



(a) Orthogonal residual of the linear prediction versus relative layer depth, one panel per task group. Solid lines show the full-sample orthogonal residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$; dashed lines show the corresponding split-half sampling-noise floor ($n/2$).



(b) Resolvability of the orthogonal residual versus relative layer depth, one panel per task group. The plotted ratio $R = \hat{s}_r^l / \phi_r^l$ is the size-matched orthogonal residual divided by the split-half sampling-noise floor; the dashed line marks the resolvability threshold ($R = 1$).

Figure 3: Orthogonal residual of the linear prediction across relative layer depth. (a) Residual magnitude for each model and task group, relative to the split-half sampling-noise floor. (b) Resolvability of the residual, measured as the size-matched residual over its noise floor.

Group	First layer			Last layer			Within floor
	full	half	floor	full	half	floor	
Arithmetic	0.38	0.45	0.47	0.76	0.81	0.75	7/40
Entity	0.34	0.68	0.94	0.67	0.69	0.45	21/40
MMLU	0.37	0.57	0.80	0.70	0.89	0.92	37/40

Table 1: Orthogonal-residual calibration via split-half consistency, at the *first* and *last* swept layer of each model, averaged over the mixed ratios and then over all five models. “full”: the *full-sample* orthogonal residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$, the minimum-variance estimate of its magnitude (0 means the mixed centroid lies exactly on the predicted point). “half”: the *size-matched* residual, a half-sample estimate at the floor’s own $n/2$ sample size. “floor”: the split-half sampling-noise floor ($n/2$). “Within floor” pools, over the five models, the swept layers (8 per model, 40 in total) whose size-matched residual is at or below the floor, i.e. indistinguishable from an exact convex combination given sampling noise. Group means can conceal real spread across models; per-model values are given in Appendix C.2 (Table 6).

The per-depth trajectory sharpens the separability-governed picture of Section 4.2. For the two well-separated groups, resolvability rises above the floor before peaking at intermediate depth. Arithmetic clears the floor earliest (relative depth ≈ 0.1) and peaks near 0.4 ($R \approx 1.5$), while Entity crosses later (≈ 0.25) but peaks higher near 0.5 ($R \approx 1.9$); both then decay toward the floor at the deepest layers. In contrast, MMLU approaches the floor monotonically rather than forming an interior peak. It exceeds the floor only marginally, and only at the deepest swept layers in Gemma-2B and Qwen-3B ($R \approx 1.05$ – 1.09), while Llama-3.2-3B and Llama-3.2-1B remain at or below the floor throughout. Thus, resolvability peaks at intermediate depth for separable task geometries but remains floor-limited for entangled ones, a distinction invisible to the nearly constant κ .

mixed centroid is indistinguishable from its exact convex-predicted point given sampling noise.

Resolvability $R_r^l = \hat{s}_r^l / \phi_r^l$ has two contributors: it increases when the higher-order orthogonal resid-

Table 2: Decomposition of the resolvability peak by task group. *Mean peak depth* is the relative layer depth at which $R_r^l = \tilde{s}_r^l / \phi_r^l$ is maximal. *Residual ROC* and *Floor ROC* are the per-model least-squares slopes of the size-matched residual $\tilde{s}_r^l$ and the noise floor ϕ_r^l against relative depth over the ascending run up to that peak, averaged across the five models (units: change per unit relative depth). A rising residual (positive residual ROC) and a falling floor (negative floor ROC) each lift resolvability; the verdict names which lever dominates.

Group	Mean peak depth	Residual ROC	Floor ROC	Verdict
Arithmetic	0.38	+1.0	0.0	Residual-driven
Entity	0.56	+0.2	-1.3	Floor-driven
MMLU	0.73	+0.5	+0.2	Floor-limited

ual grows (numerator up) and when task separability sharpens, shrinking the sampling-noise floor (denominator down; Section 4.2). To determine which dominates each task group, we decompose the ascending portion of R_r^l up to its peak by measuring, for each model, the least-squares slopes of the residual and floor against relative depth, then averaging across models (Table 2).

Arithmetic is *residual-driven*: the residual rises steeply (mean slope $\approx +1.0$) while the floor remains nearly constant (≈ 0.0), producing an early peak (≈ 0.38). Entity is *floor-driven*: a nearly flat residual ($\approx +0.2$) is revealed by a rapidly collapsing floor (≈ -1.3), yielding a later and higher peak (≈ 0.56). MMLU is *floor-limited*: although its residual grows ($\approx +0.5$), its floor does not fall ($\approx +0.2$), so resolvability never develops an interior peak and remains near the sampling floor, exceeding it only marginally in Gemma-2B and Qwen-3B. Thus, the orthogonal component becomes resolvable only when residual growth is accompanied by increasing task separability.

4.4 Ablating the systematic orthogonal residual

We now ask whether the orthogonal residual is behaviorally load-bearing, by removing it from the residual stream and measuring the effect on the model’s output.

Stage 1: extract the residual. For each mixed ratio r and layer l we average the contrast vectors over that ratio’s held-out questions to get its mean contrast $\bar{\mathbf{c}}_r^l$, and keep the part of its displacement from the tri-center μ^l that the convex prediction does not explain:

$$\mathbf{v}_\perp(r) := (\bar{\mathbf{c}}_r^l - \mu^l) - \kappa(r) \hat{\mathbf{d}}_r^l \perp \hat{\mathbf{d}}_r^l,$$

where $\hat{\mathbf{d}}_r^l$ is the direction of the convex prediction for ratio r and $\kappa(r)$ is the projection of the displacement onto it. So $\mathbf{v}_\perp(r)$ is the *average orthogonal residual of a ratio*: one vector per ratio and layer, not per question. Averaging over questions cancels the idiosyncrasy of any individual prompt, and what survives is the component shared across questions at that ratio—a property of the task ratio itself, and the object on which a claim about *task* composition should be tested. The questions used here are disjoint from those the intervention is evaluated on: a mean taken over the ablated questions would contain a $1/n$ share of each question’s own residual, handing the ablation an in-sample advantage that a random direction does not enjoy.

Stage 2: remove it. For each evaluated question we subtract $\mathbf{v}_\perp(r)$ from that question’s own few-shot activation at the same ratio r and the same layer l , at the assistant-boundary state—so a ratio’s residual is only ever removed from that ratio’s prompts, at the layer it was fit. The null arm substitutes one of $D = 30$ random vectors $\mathbf{g}$, each also orthogonal to $\hat{\mathbf{d}}_r^l$ and rescaled so that $\|\mathbf{g}\| = \|\mathbf{v}_\perp(r)\|$. Both arms remove a single out-of-sample vector from the few-shot residual stream, and differ only in its direction.

Stage 3: measure the shift. We report $\text{KL}(\text{few-shot} \parallel \text{ablated})$ averaged over the $n = 10$ evaluation questions, and the ablation’s rank among its 30 draws; with $D = 30$ the smallest attainable one-sided p -value is $1/31 \approx 0.032$ (Phipson and Smyth, 2010). Because the two arms are matched in norm, in orthogonality, and in the prompts and questions they act on, any gap between them is attributable to direction alone. The question split, construction details, the faithfulness control, and per-layer values are deferred to Appendix D.

The residual is small. Its relative magnitude $\rho := \|\mathbf{v}_\perp(r)\| / \|\mathbf{h}_l\|$ grows with depth but stays a small fraction of the ambient activation, rising from 0.003–0.012 at the shallowest probed layer to 0.162 (Arithmetic), 0.242 (Entity) and 0.067 (MMLU) at three-quarter depth (Table 7), while κ stays within $[0.85, 1.09]$ across all groups and depths. The mean displacement is, to a good approximation, the convex prediction itself—consistent with the $\kappa \approx 1$ result of Section 3.3, but now stated for the vector that remains once the prediction is subtracted.

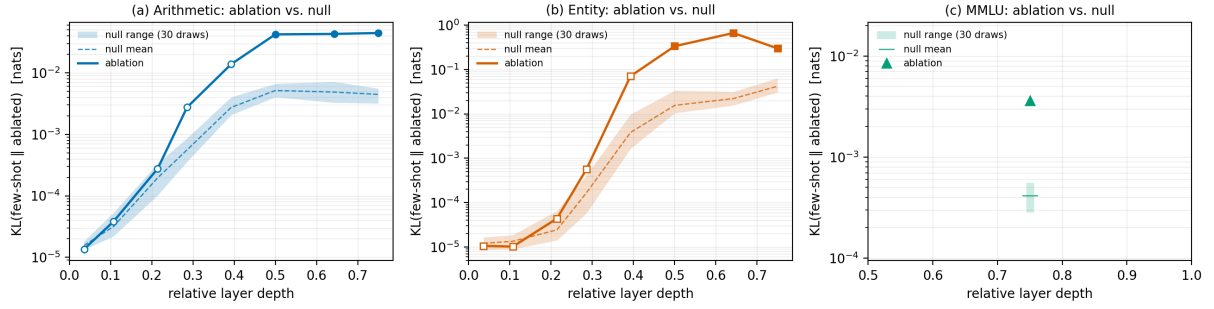


Figure 4: Ablating the average orthogonal residual $\mathbf{v}_\perp(r)$ at the assistant-boundary token (Llama-3.2-3B), against relative layer depth, one panel per task group: **(a)** Arithmetic, **(b)** Entity, **(c)** MMLU. Each shows the KL divergence of the ablated next-token distribution from the few-shot distribution (solid), against the norm-matched random-orthogonal null (dashed: null mean; band: full range of the $D = 30$ draws). Filled markers denote layers where the ablation exceeds every null draw in every mixed ratio; hollow markers denote layers where it does not. MMLU was run at layer 21 only, so it appears as a single point rather than a curve. Points are means over 10 evaluation questions and 6 mixed ratios, with centroids fit on 40 disjoint questions (20 for MMLU). Note the log scale, and that the three panels use different y -ranges. Per-layer residual magnitudes ρ are given in Table 7.

Its direction is load-bearing. At the deepest layer we probe (layer 21, three-quarter depth), removing $\mathbf{v}_\perp(r)$ shifts the next-token distribution more than *every* one of the 30 norm-matched random orthogonal draws, in *all six* mixed ratios, in *all three* task groups ($p \leq 0.032$ per ratio; Figure 4). Pooled over ratios, the ablation reaches 0.044 nats against a null mean of 0.005 for Arithmetic— $8.0\times$ its largest draw—0.297 against 0.042 for Entity ($4.7\times$ its largest draw), and 0.0036 against 0.0004 for MMLU ($6.5\times$ its largest draw). Removing this particular orthogonal direction therefore costs more than removing any other orthogonal direction of the same length. What the convex prediction misses is not just a perturbation of a certain size; it points somewhere in particular, and that direction carries the effect. For the two groups with full depth coverage, the effect is confined to the deeper half of the network: it clears every null draw at every depth ≥ 0.50 , while at the shallowest probed layer it lands wherever a random direction of the same length would, its percentile scattering from the 3rd to the 87th across ratios. This shallow regime is a built-in negative control—before the boundary token has aggregated the in-context examples there is no systematic residual to remove, and the test correctly fails to separate. That the same pipeline returns a null result where one is expected and a maximal one where it is predicted is the strongest internal evidence that the deep effect is not an artifact of patching.

MMLU: magnitude and direction dissociate. Although MMLU has the smallest residual and is statistically unresolved under the split-half test,

its held-out residual direction consistently outperforms norm-matched random directions. This is not contradictory because the two tests answer different questions.

5 Conclusion

The convex-combination prediction holds to first order at every depth we probe, but it is flat across layers: what changes with depth is the *orthogonal* residual that prediction ignores. The residual grows steadily with depth, yet whether it is statistically resolvable is governed by task separability rather than by its own magnitude—so it resolves in the middle layers, and in the most weakly separated task group scarcely at all. It is load-bearing regardless: removing a ratio’s average orthogonal residual, estimated on held-out questions, perturbs the next-token distribution more than any norm-matched random orthogonal direction, so it is the residual’s *direction* rather than the mere fact of perturbing the activation that carries the effect. Magnitude and resolvability are therefore poor guides to whether a direction matters behaviorally: the first-order linear description is scale-accurate, but the structure it discards is real, systematic, and causally active.

Limitations

Positional encoding. We do not directly control for positional encoding, although we are able to demonstrate linearity without explicitly controlling for this factor.

Model scale and coverage. Model scale is capped at 8B parameters due to hardware

constraints, so whether the same depth- and separability-resolved picture holds at larger scales is left open.

Single model for causal ablation. The residual-ablation intervention (Section 4.4) is run on a single model, Llama-3.2-3B, picked arbitrarily among those tested due to hardware constraints; whether the same causal picture holds at other model families and scales is left open.

Depth coverage of the MMLU null. The norm-matched null of Section 4.4 is reported at every swept layer for Arithmetic and Entity, but at layer 21 (three-quarter depth) only for MMLU: its few-shot prompts are $\approx 30\times$ longer than the other groups’ (three full question-and-options exemplars, $\approx 1,000$ tokens on average versus ≈ 30), and a full sweep of the $\approx 1.5 \times 10^4$ patched forward passes it requires did not fit the compute budget available to us. The headline claim is therefore read at layer 21, where all three groups have data. Two consequences: MMLU contributes no shallow-layer negative control of its own, so the evidence that the effect is depth-dependent rather than a patching artifact rests on Arithmetic and Entity; and MMLU’s centroids are fit on 20 held-out questions rather than 40, the pool being smaller, making its $\mathbf{v}_\perp(r)$ the noisiest of the three—which, if anything, works against the result it obtains.

Depth coverage. Our layer sweep extends only to a relative depth of 0.75, leaving the final $\sim 25\%$ of layers unprobed. This is consistent with the emerging picture of how depth is used: the last stage of inference is a “residual sharpening” phase that suppresses features to finalize the next-token distribution rather than building the representations we analyze (Lad et al., 2025), and intermediate rather than final layers carry the richest, most linearly structured task representations (Skean et al., 2024). The task-composition geometry we study is therefore expected to be established by this depth; we nonetheless do not verify that the linear structure persists—or degrades in the manner these accounts predict—across the terminal output-oriented layers, and leave a targeted sweep of that band to future work.

References

- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, and 12 others. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.
- Yuxin Dong, Jiachen Jiang, Zhihui Zhu, and Xia Ning. 2026. Understanding task vectors in in-context learning: Emergence, functionality, and limitations. In *International Conference on Learning Representations*.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. 2022. [Toy models of superposition](#). *Transformer Circuits Thread*.
- Gemma Team. 2024. [Gemma 2: Improving open language models at a practical size](#). *Preprint*, arXiv:2408.00118.
- Aaron Grattafiori and 1 others. 2024. [The Llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.
- Roe Hendel, Mor Geva, and Amir Globerson. 2023. In-context learning creates task vectors. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9318–9333, Singapore. Association for Computational Linguistics.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *International Conference on Learning Representations*.
- Jiachen Jiang, Yuxin Dong, Jinxin Zhou, and Zhihui Zhu. 2025. [From compression to expression: A layerwise analysis of in-context learning](#). *Preprint*, arXiv:2505.17322.
- Vedang Lad, Jin Hwa Lee, Wes Gurnee, and Max Tegmark. 2025. [The remarkable robustness of llms: Stages of inference?](#) *Preprint*, arXiv:2406.19384.
- Belinda Phipson and Gordon K Smyth. 2010. [Permutation p-values should never be zero: Calculating exact p-values when permutations are randomly drawn](#). *Statistical Applications in Genetics and Molecular Biology*, 9(1):Article 39.
- Oscar Skean, Md Rifat Arefin, Yann LeCun, and Ravid Shwartz-Ziv. 2024. [Does representation matter? exploring intermediate layers in large language models](#). *Preprint*, arXiv:2412.09563.
- Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. 2024. Function vectors in large language models. In *International Conference on Learning Representations*.
- Zheyang Xiong, Ziyang Cai, John Cooper, Albert Ge, Vasilis Papageorgiou, Zack Sifakis, Angeliki Gian-nou, Ziqian Lin, Liu Yang, Saurabh Agarwal, Grigorios G Chrysos, Samet Oymak, Kangwook Lee, and Dimitris Papailiopoulos. 2025. Everything everywhere all at once: LLMs can in-context learn multiple tasks in superposition. In *Proceedings of the 42nd International Conference on Machine Learning*.
- Hao Yan, Haolin Yang, and Yiqiao Zhong. 2026. [Task vector geometry underlies dual modes of task inference in transformers](#). *Preprint*, arXiv:2605.03780.
- An Yang and 1 others. 2024. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.

A Implementation details for empirical verification

A.1 Prompt construction

For each of the 10 ratios listed in Section 3.1, n samples of in-context examples and test questions are created for each task group. $n = 50$ for the Arithmetic and Entity task groups and $n = 30$ for MMLU tasks.

For arithmetic tasks, op1 and op2 are sampled from the range $[1, 49]$ and are formatted as Direct, MCQ, or Verification tasks. For entity tasks, countries are sampled and their corresponding entities are extracted based on the ratio. For MMLU tasks, questions for a sample are extracted from each task (subject/domain) based on the ratio.

For a given task group and ratio, each sample constitutes (1) the in context examples concatenated with the test question to produce the few shot prompt, and (2) the corresponding test question, used as the zero-shot baseline. For all three task groups, the test question is extracted from a heldout pool disjoint from the in-context examples. Implementation details on sampling the ICL examples and test question can be found in Appendix E.

A.2 Layer selection

We use the following relative layer depths for each model: $[0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.65, 0.75]$. The higher density in earlier layers is intended to capture the region where task-relevant information is most concentrated, motivated by findings that task and function vectors are most effective when injected in the early-to-middle layers of autoregressive LLMs (Hendel et al., 2023; Todd et al., 2024).

A.3 Contrast vector extraction

For each sample, we first collect the activations at the final token of the residual stream at the assistant boundary turn. Since these activations are independent of the task ratio in in-context examples, they are precomputed and stored in a temporary cache for efficiency. This is done by rotating the task corresponding to the test question for each sample. That is, for sample index i , the test question is drawn from the task $\text{TASKS}[i \% \text{num_tasks}]$.

Next, for each ratio, we conduct another forward pass to collect the activations at the assistant header when given the few shot prompts and the corresponding test questions used in the zero shot prompts.

The stored zero shot activations are then subtracted from the corresponding few shot activations to produce contrast vectors at each layer of the model for each ratio. For visualizing these vectors, following the method used by Xiong et al. (2025), we use LDA projection. LDA is fit on the pure condition contrast vectors only $((3, 0, 0), (0, 3, 0), \text{ and } (0, 0, 3))$. This is then used to transform the contrast vectors from all 10 ratios. A separate LDA is fit per layer per task group.

A.4 Calculating κ

The above contrast vectors are then used to compute μ_r^l , $\hat{\mu}_r^l$, and μ^l as described in Section 3.3 for each layer at each ratio. The calculated values of κ are then stored per model and task group. κ is calculated using the original contrast vectors, not the LDA projected vectors, providing a measurement of linearity in the original activation space.

Further, as stated in Section 3.3, κ is not well defined for the 1-1-1 ratio and is therefore not computed.

B Calibration and aggregate statistics for κ

B.1 Exact permutation null

To calibrate the observed κ values, we use an exact permutation null. κ is defined on the $K = 6$ mixed ratios whose predicted displacement $\hat{\mathbf{d}}_r^l$ is nonzero (the three pure ratios define the basis, and 1-1-1 is undefined). The observed statistic at layer l is the mean $S := \frac{1}{K} \sum_r \kappa_r^l$. A permutation π re-pairs each

observed displacement $\mathbf{d}_r^l$ with a permuted predicted displacement $\hat{\mathbf{d}}_{\pi(r)}^l$, yielding

$$S(\pi) := \frac{1}{K} \sum_r \frac{\mathbf{d}_r^l \cdot \hat{\mathbf{d}}_{\pi(r)}^l}{\|\hat{\mathbf{d}}_{\pi(r)}^l\|^2}$$

We enumerate all $6! = 720$ permutations and report the one-sided $p = \#\{\pi : S(\pi) \geq S\}/720$. The identity permutation is included, so the smallest attainable value is $p = 1/720 \approx 1.4 \times 10^{-3}$; this is the exact-enumeration analogue of the add-one correction (Phipson and Smyth, 2010). The null is computed per layer and stored alongside κ for each model and task group.

B.2 Aggregate statistics across ratios and depth

Tables 3–5 report the full κ distribution behind the near-constant summary quoted in the main text. For each task group–model pair and each swept layer l (relative depth $d = l/L$), we give the mean $\bar{\kappa}$, standard deviation σ_κ , and range $[\min, \max]$ of κ_r^l over the $K = 6$ mixed ratios. The per-layer mean stays close to 1 at every depth in every cell, confirming that linear superposition is scale-accurate throughout; the dispersion across ratios widens somewhat at the deepest layers—most visibly in the Arithmetic group, where individual ratios spread while the mean holds near 1—consistent with the growing orthogonal residual analyzed in Section 4.2 and Appendix C.

Table 3: Aggregate κ statistics for the Arithmetic task group: per-layer mean, standard deviation, and range across the $K = 6$ mixed ratios, for each model. Relative depth is $d = l/L$.

Model	$l(d)$	$\bar{\kappa}$	σ_κ	range
Llama 3.2 1B	1 (0.06)	1.01	0.16	[0.76, 1.25]
	2 (0.12)	1.09	0.26	[0.68, 1.42]
	3 (0.19)	1.11	0.23	[0.73, 1.33]
	5 (0.31)	1.10	0.14	[0.99, 1.30]
	6 (0.38)	1.07	0.23	[0.72, 1.33]
	8 (0.50)	0.97	0.18	[0.83, 1.31]
	10 (0.62)	1.01	0.49	[0.35, 1.81]
	12 (0.75)	1.03	0.64	[0.15, 1.94]
Gemma 2 2B	1 (0.04)	1.11	0.24	[0.77, 1.40]
	3 (0.12)	1.06	0.25	[0.64, 1.26]
	5 (0.19)	1.09	0.28	[0.64, 1.44]
	8 (0.31)	1.06	0.14	[0.80, 1.16]
	10 (0.38)	1.03	0.14	[0.80, 1.18]
	13 (0.50)	0.86	0.33	[0.36, 1.20]
	17 (0.65)	0.92	0.31	[0.64, 1.36]
	20 (0.77)	0.97	0.30	[0.73, 1.42]
Llama 3.2 3B	1 (0.04)	1.01	0.14	[0.76, 1.17]
	3 (0.11)	1.06	0.16	[0.80, 1.22]
	6 (0.21)	1.08	0.20	[0.83, 1.39]
	8 (0.29)	1.09	0.21	[0.85, 1.36]
	11 (0.39)	1.07	0.26	[0.76, 1.37]
	14 (0.50)	0.91	0.44	[0.33, 1.44]
	18 (0.64)	0.88	0.53	[0.01, 1.41]
	21 (0.75)	0.86	0.60	[-0.18, 1.43]
Qwen 2.5 3B	2 (0.06)	1.01	0.23	[0.63, 1.31]
	4 (0.11)	1.05	0.15	[0.80, 1.22]
	7 (0.19)	1.04	0.21	[0.68, 1.20]
	11 (0.31)	1.05	0.26	[0.72, 1.35]
	14 (0.39)	1.08	0.28	[0.81, 1.57]
	18 (0.50)	1.07	0.33	[0.75, 1.63]
	23 (0.64)	1.05	0.22	[0.74, 1.40]
	27 (0.75)	1.06	0.09	[0.92, 1.14]
Llama 3.1 8B	2 (0.06)	1.05	0.12	[0.83, 1.14]
	3 (0.09)	1.04	0.13	[0.82, 1.14]
	6 (0.19)	1.07	0.14	[0.88, 1.29]
	10 (0.31)	1.04	0.13	[0.88, 1.22]
	13 (0.41)	0.97	0.28	[0.74, 1.32]
	16 (0.50)	0.91	0.31	[0.45, 1.28]
	21 (0.66)	0.94	0.52	[0.28, 1.50]
	24 (0.75)	0.95	0.60	[0.23, 1.61]

Table 4: Aggregate κ statistics for the Entity task group: per-layer mean, standard deviation, and range across the $K = 6$ mixed ratios, for each model. Relative depth is $d = l/L$.

Model	l (d)	$\bar{\kappa}$	σ_{κ}	range
Llama 3.2 1B	1 (0.06)	1.04	0.07	[0.94, 1.12]
	2 (0.12)	1.02	0.05	[0.94, 1.10]
	3 (0.19)	1.02	0.04	[0.96, 1.06]
	5 (0.31)	0.97	0.11	[0.84, 1.08]
	6 (0.38)	0.97	0.05	[0.90, 1.02]
	8 (0.50)	0.97	0.13	[0.73, 1.10]
	10 (0.62)	0.94	0.20	[0.68, 1.17]
	12 (0.75)	0.93	0.28	[0.61, 1.22]
Gemma 2 2B	1 (0.04)	1.02	0.05	[0.93, 1.06]
	3 (0.12)	1.00	0.03	[0.96, 1.03]
	5 (0.19)	1.00	0.07	[0.89, 1.08]
	8 (0.31)	1.05	0.17	[0.81, 1.22]
	10 (0.38)	1.03	0.15	[0.88, 1.26]
	13 (0.50)	1.03	0.31	[0.60, 1.43]
	17 (0.65)	1.04	0.22	[0.80, 1.32]
	20 (0.77)	1.14	0.19	[0.87, 1.30]
Llama 3.2 3B	1 (0.04)	1.04	0.09	[0.90, 1.15]
	3 (0.11)	1.02	0.06	[0.95, 1.11]
	6 (0.21)	1.03	0.11	[0.87, 1.14]
	8 (0.29)	0.99	0.09	[0.90, 1.13]
	11 (0.39)	1.00	0.43	[0.20, 1.37]
	14 (0.50)	0.91	0.30	[0.51, 1.30]
	18 (0.64)	1.03	0.22	[0.70, 1.27]
	21 (0.75)	1.10	0.32	[0.61, 1.47]
Qwen 2.5 3B	2 (0.06)	1.08	0.08	[0.99, 1.21]
	4 (0.11)	1.08	0.15	[0.84, 1.26]
	7 (0.19)	1.04	0.11	[0.85, 1.15]
	11 (0.31)	0.99	0.16	[0.78, 1.22]
	14 (0.39)	0.97	0.16	[0.72, 1.19]
	18 (0.50)	0.99	0.14	[0.88, 1.24]
	23 (0.64)	1.06	0.34	[0.63, 1.52]
	27 (0.75)	1.06	0.31	[0.57, 1.55]
Llama 3.1 8B	2 (0.06)	1.06	0.09	[0.93, 1.18]
	3 (0.09)	1.04	0.07	[0.96, 1.13]
	6 (0.19)	1.03	0.12	[0.84, 1.17]
	10 (0.31)	0.97	0.09	[0.90, 1.12]
	13 (0.41)	0.89	0.21	[0.64, 1.22]
	16 (0.50)	0.75	0.27	[0.36, 1.04]
	21 (0.66)	0.82	0.28	[0.53, 1.14]
	24 (0.75)	0.87	0.31	[0.51, 1.31]

Table 5: Aggregate κ statistics for the MMLU task group: per-layer mean, standard deviation, and range across the $K = 6$ mixed ratios, for each model. Relative depth is $d = l/L$.

Model	l (d)	$\bar{\kappa}$	σ_{κ}	range
Llama 3.2 1B	1 (0.06)	0.97	0.24	[0.68, 1.36]
	2 (0.12)	0.97	0.25	[0.75, 1.39]
	3 (0.19)	1.02	0.15	[0.79, 1.19]
	5 (0.31)	0.98	0.19	[0.72, 1.20]
	6 (0.38)	0.98	0.15	[0.75, 1.16]
	8 (0.50)	0.98	0.21	[0.72, 1.18]
	10 (0.62)	0.99	0.23	[0.70, 1.19]
	12 (0.75)	0.94	0.31	[0.56, 1.22]
Gemma 2 2B	1 (0.04)	0.97	0.19	[0.65, 1.21]
	3 (0.12)	1.00	0.27	[0.65, 1.46]
	5 (0.19)	0.99	0.22	[0.71, 1.37]
	8 (0.31)	0.95	0.07	[0.84, 1.01]
	10 (0.38)	0.96	0.10	[0.82, 1.07]
	13 (0.50)	0.98	0.11	[0.80, 1.12]
	17 (0.65)	1.01	0.31	[0.50, 1.32]
	20 (0.77)	0.99	0.59	[0.14, 1.63]
Llama 3.2 3B	1 (0.04)	0.97	0.25	[0.65, 1.38]
	3 (0.11)	0.99	0.10	[0.91, 1.15]
	6 (0.21)	1.01	0.07	[0.92, 1.10]
	8 (0.29)	1.02	0.08	[0.90, 1.11]
	11 (0.39)	1.03	0.12	[0.90, 1.23]
	14 (0.50)	1.00	0.15	[0.79, 1.22]
	18 (0.64)	1.00	0.06	[0.92, 1.08]
	21 (0.75)	1.03	0.05	[0.99, 1.11]
Qwen 2.5 3B	2 (0.06)	0.99	0.20	[0.71, 1.29]
	4 (0.11)	0.99	0.18	[0.77, 1.28]
	7 (0.19)	0.99	0.11	[0.81, 1.11]
	11 (0.31)	0.99	0.09	[0.83, 1.07]
	14 (0.39)	0.96	0.12	[0.75, 1.06]
	18 (0.50)	0.94	0.14	[0.76, 1.09]
	23 (0.64)	0.94	0.08	[0.81, 1.01]
	27 (0.75)	0.91	0.07	[0.78, 0.98]
Llama 3.1 8B	2 (0.06)	0.98	0.22	[0.69, 1.32]
	3 (0.09)	0.98	0.16	[0.79, 1.25]
	6 (0.19)	0.99	0.08	[0.93, 1.12]
	10 (0.31)	1.01	0.07	[0.93, 1.10]
	13 (0.41)	1.03	0.07	[0.94, 1.12]
	16 (0.50)	1.00	0.04	[0.93, 1.04]
	21 (0.66)	1.00	0.08	[0.87, 1.10]
	24 (0.75)	1.01	0.11	[0.88, 1.13]

C Split-half noise floor and per-model residual calibration

C.1 Construction of the floor

κ measures the component of the observed displacement $\mathbf{d}_r^l$ that lies *along* the predicted displacement $\hat{\mathbf{d}}_r^l$; the complementary *orthogonal* residual is captured by $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$, the fraction of $\mathbf{d}_r^l$ pointing off the predicted direction ($\sin \theta = 0$ iff the mixed centroid lies exactly on the predicted point). In finite samples this residual cannot reach 0, since $\mathbf{d}_r^l$ is an empirical centroid carrying sampling noise; comparing it to 0 is therefore uninformative. We instead calibrate it against the best alignment the data can resolve. For each mixed ratio r we randomly partition its n samples into two halves and form a displacement $\mathbf{d}^{(1)}, \mathbf{d}^{(2)}$ from each half, holding the pure-task centroids—and hence the triangle center and the prediction basis—at their full-sample values so that only the sampling noise in μ_r varies. The noise floor is $\sin \theta(\mathbf{d}^{(1)}, \mathbf{d}^{(2)})$, averaged over 500 random splits: no prediction can be expected to align with $\mathbf{d}_r^l$ more closely than an independent estimate of $\mathbf{d}_r^l$ aligns with itself. This yields two residual estimates that serve two distinct purposes. To report the *magnitude* of the orthogonal component and its trend across depth, we use the full-sample residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$ computed from all n samples: it is the minimum-variance estimate, so it tracks the true magnitude most cleanly and is not flattened by sampling noise (this is the quantity plotted in Figure 3a and reported in the “full” columns of Tables 1 and 6). To *test resolvability*—whether the residual exceeds what sampling noise alone would produce—we instead form a size-matched residual

on the same $n/2$ half-sample displacements as the floor, since these angles shrink with sample size and only an equal- n comparison is fair. The reported “within floor” flag is this size-matched residual falling at or below the floor: a residual within the floor indicates the convex prediction is as accurate as sampling noise permits, while one above it indicates a statistically resolvable orthogonal component. Reporting the full-sample residual against the $n/2$ floor would be a more lenient, best-case comparison, so we reserve the floor strictly for the size-matched test. The floor is a normalized, scale-free quantity, so it is comparable across layers, models, and task groups despite differing activation magnitudes.

C.2 Per-model values

Table 6 gives the per-model breakdown behind the group means of Table 1, with each group’s mean repeated in italics for reference. The group-level ordering—Arithmetic’s residual resolvable almost everywhere, Entity’s intermediate, MMLU’s within the floor at nearly every layer—is never reversed in an individual model, and is strict in four of the five; the exception is Gemma 2 2B, where Arithmetic and Entity tie at 3/8 within-floor layers. The means of Table 1 are therefore not artifacts of averaging over a heterogeneous set. The spread the means conceal is nonetheless real and is concentrated in two places. The deepest-layer floor varies substantially within the Arithmetic group (0.69–0.86), which is what drives the differing within-floor counts there (0/8 for Qwen 2.5 3B and Llama 3.1 8B, against 3/8 for Gemma 2 2B); and MMLU’s deepest-layer full-sample residual ranges from 0.53 (Llama 3.2 3B) to 0.83 (Llama 3.1 8B) around a mean of 0.70. Entity is the most homogeneous group, its deepest-layer floor sitting at 0.43 in four of the five models.

Group	Model	First layer			Last layer			Within floor
		full	half	floor	full	half	floor	
Arithmetic	Llama 3.2 1B	0.33	0.43	0.53	0.78	0.81	0.71	2/8
	Gemma 2 2B	0.36	0.44	0.49	0.74	0.81	0.86	3/8
	Llama 3.2 3B	0.41	0.49	0.51	0.75	0.79	0.69	2/8
	Qwen 2.5 3B	0.40	0.44	0.39	0.71	0.76	0.73	0/8
	Llama 3.1 8B	0.42	0.47	0.43	0.82	0.85	0.75	0/8
<i>Mean</i>		<i>0.38</i>	<i>0.45</i>	<i>0.47</i>	<i>0.76</i>	<i>0.81</i>	<i>0.75</i>	<i>7/40</i>
Entity	Llama 3.2 1B	0.32	0.70	0.96	0.62	0.65	0.52	5/8
	Gemma 2 2B	0.34	0.76	0.97	0.56	0.59	0.43	3/8
	Llama 3.2 3B	0.33	0.65	0.94	0.66	0.67	0.43	4/8
	Qwen 2.5 3B	0.34	0.61	0.89	0.69	0.71	0.43	6/8
	Llama 3.1 8B	0.39	0.66	0.93	0.80	0.81	0.43	3/8
<i>Mean</i>		<i>0.34</i>	<i>0.68</i>	<i>0.94</i>	<i>0.67</i>	<i>0.69</i>	<i>0.45</i>	<i>21/40</i>
MMLU	Llama 3.2 1B	0.40	0.61	0.83	0.70	0.87	0.98	8/8
	Gemma 2 2B	0.36	0.56	0.79	0.72	0.91	0.87	6/8
	Llama 3.2 3B	0.40	0.59	0.81	0.53	0.80	0.97	8/8
	Qwen 2.5 3B	0.30	0.48	0.72	0.70	0.93	0.85	7/8
	Llama 3.1 8B	0.38	0.59	0.83	0.83	0.93	0.95	8/8
<i>Mean</i>		<i>0.37</i>	<i>0.57</i>	<i>0.80</i>	<i>0.70</i>	<i>0.89</i>	<i>0.92</i>	<i>37/40</i>

Table 6: Orthogonal-residual calibration via split-half consistency, per model and task group—the full breakdown summarized in Table 1. Columns are as in Table 1: for the *first* and *last* swept layer, averaged over the mixed ratios, the full-sample residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$ (“full”), the size-matched $n/2$ residual (“half”), and the split-half sampling-noise floor (“floor”). “Within floor” counts the swept layers (of 8) whose size-matched residual is at or below the floor. Italicized *Mean* rows are the across-model means reported in Table 1; the within-floor mean is pooled over all 40 model–layer pairs rather than averaged.

D Residual ablation details

Decomposition. For a mixed ratio r at layer l , let $\bar{\mathbf{c}}_r^l = \frac{1}{n} \sum_i \mathbf{c}_r^l(x_i)$ be the mean contrast over the estimation questions and μ^l the tri-center (the mean of the three pure-ratio mean contrasts). Writing $\mathbf{d} = \bar{\mathbf{c}}_r^l - \mu^l$ for the displacement and $\hat{\mathbf{d}}_r^l$ for the convex prediction’s direction, the projection coefficient

Layer	Depth	Arithmetic			Entity			MMLU		
		ρ	KL	null	ρ	KL	null	ρ	KL	null
1	0.04	0.011	0.0000	0.0000	0.003	0.0000	0.0000	0.012	—	—
3	0.11	0.033	0.0000	0.0000	0.006	0.0000	0.0000	0.025	—	—
6	0.21	0.068	0.0003	0.0002	0.012	0.0000	0.0000	0.043	—	—
8	0.29	0.098	0.0028	0.0006	0.036	0.0006	0.0002	0.055	—	—
11	0.39	0.142	0.0138	0.0027	0.116	0.0705	0.0038	0.071	—	—
14	0.50	0.169	0.0422*	0.0052	0.252	0.3336*	0.0155	0.069	—	—
18	0.64	0.169	0.0429*	0.0049	0.229	0.6604*	0.0221	0.065	—	—
21	0.75	0.162	0.0444*	0.0045	0.242	0.2974*	0.0418	0.067	0.0036*	0.0004

Table 7: Per-layer average residual magnitude $\rho = \|\mathbf{v}_\perp(r)\|/\|\mathbf{h}\|$, ablation divergence KL(few-shot $\parallel$ ablated) (nats), and the mean of the $D = 30$ norm-matched random-orthogonal null, for Llama-3.2-3B, averaged over 10 evaluation questions and the 6 mixed ratios, with centroids fit on 40 disjoint questions (20 for MMLU, whose pool is smaller). *: the ablation exceeds all 30 null draws in every mixed ratio at that layer ($p \leq 1/31 \approx 0.032$ per cell); the claim of Section 4.4 is read at layer 21, the only layer at which all three groups have a null. MMLU’s null was run at layer 21 only (—: not run); its ρ column is pure geometry and is available at every layer.

and the residual of Section 4.4 are

$$\kappa(r) = \frac{\langle \mathbf{d}, \hat{\mathbf{d}}_r^l \rangle}{\langle \hat{\mathbf{d}}_r^l, \hat{\mathbf{d}}_r^l \rangle}, \quad \mathbf{v}_\perp(r) = \mathbf{d} - \kappa(r) \hat{\mathbf{d}}_r^l,$$

so that $\mathbf{v}_\perp(r) \perp \hat{\mathbf{d}}_r^l$ by construction.

Faithfulness control. Overwriting the boundary state with the intact $\mathbf{h}_l(p_r(x_i))$ is a no-op: across all layers, ratios, and test questions the patched logits match the clean few-shot logits with $\max |\Delta \text{logit}| = 0$ and $\text{KL} = 0$, verifying that the patching mechanism itself introduces no artifact. All divergences reported are therefore attributable to the removal of $\mathbf{v}_\perp(r)$.

Null construction. Each $\mathbf{g}$ is drawn i.i.d. from $\mathcal{N}(0, I)$, projected onto the orthogonal complement of $\hat{\mathbf{d}}_r^l$, normalized, and rescaled to $\|\mathbf{v}_\perp(r)\|$ —so the null inherits its displacement magnitude from the held-out residual. The realized alignment with the true residual is negligible ($|\cos(\mathbf{g}, \mathbf{v}_\perp(r))| = 0.013\text{--}0.015$), confirming the draws do not accidentally recover the ablated direction. The D draws for a given (r, l) are evaluated as one batched forward pass over the shared prompt; batch rows do not interact, so this is arithmetically identical to D separate passes.

Protocol. We use $n = 10$ evaluation questions per ratio and the six non-degenerate mixed ratios (the balanced 1-1-1 ratio is excluded as degenerate: its convex prediction coincides with the tri-center, so $\hat{\mathbf{d}}$ is undefined), at layers taken at fixed fractions of model depth. Each ratio’s question pool is split disjointly by index: questions 0–9 are the $n = 10$ on which the intervention is evaluated, and every fitted quantity—pure-ratio mean contrasts, tri-center, and the mixed-ratio mean contrast—is estimated from the remainder, so no question contributes to both the estimation of $\mathbf{v}_\perp(r)$ and its evaluation. Arithmetic and Entity draw on a pool of 50 questions per ratio, leaving 40 for estimation; MMLU’s pool is 30, leaving 20, so its $\mathbf{v}_\perp(r)$ is the noisiest of the three. The ambient norm $\|\mathbf{h}\|$ is that of the actual few-shot boundary activation. Both ρ and κ are reported as means over the six mixed ratios. Per-layer values are given in Table 7.

E Implementation details for task group sampling

E.1 Prompt templates

Table 8 contains the formats for each task within the three groups, as well as examples of an in-context question-answer pair and a test question.

E.2 Arithmetic

First, a pool of 200 in-context and 50 test pairs (op1, op2), with each op1, op2 $\in [1, 49]$, are generated using seeds 42 and 43 respectively. Duplicate pairs are re-generated until the ICL and test pools are

Table 8: Prompt formats for each task and task group.

Group	Task	In-context Example	Test Question
Arithmetic	Direct	User: {op1} + {op2} A: Assistant: {op1+op2}	User: {op1} + {op2} A:
	MCQ	User: {op1}+{op2}=? A) {optA} B) {optB} C) {optC} D) {optD} A: Assistant: B	User: {op1}+{op2}=? A) {optA} B) {optB} C) {optC} D) {optD} A:
	Verification	User: {op1}+{op2}=x. Correct? A: Assistant: yes/no	User: {op1}+{op2}=x. Correct? A:
Entity	Capital	User: {country} Assistant: capital[{country}]	User: {country}
	Currency	User: {country} Assistant: currency[{country}]	User: {country}
	Language	User: {country} Assistant: language[{country}]	User: {country}
MMLU	History/Law/ML	User: Question: {question} A) {optA} B) {optB} C) {optC} D) {optD} Answer: Assistant: B	User: Question: {question} A) {optA} B) {optB} C) {optC} D) {optD} Answer:

disjoint. Thus, while the same (op1, op2) pair can appear in more than 1 in-context example, the in-context and test pools are disjoint. For each sample using ratio r of task combinations, 3 in-context op pairs and 1 test op pair are drawn without replacement. Each of the ICL pairs are formatted according to the ratio r and are concatenated with the test op pair.

Generating distractors for multiple choice: the distractors are within 10 of the true value of $op1 + op2$. This is to prevent lexical differences in the options having an unintended influence on the intended task to represent. For each ratio r , we use $n = 50$ samples.

E.3 Entity

We manually curate a dataset of 96 countries and map each to its capital, currency, and language. This dataset is partitioned into an in-context pool of 66 countries and a test pool of 30 countries, making the two pools disjoint.

To generate the in-context examples for a prompt, we sample countries without replacement from the in-context pool for each task according to the ratio r . This guarantees that countries are not repeated for a single task (that is, France $\rightarrow$ Euro does not appear twice in one sample prompt), but allows for cross-task duplication of countries (that is, France $\rightarrow$ Euro and France $\rightarrow$ Paris can appear in the same prompt). The test question is separately drawn from the test pool.

We use $n = 50$ samples for each in-context ratio. We do not add an assistant trigger in the user prompt after the test question; the prompt format is consistent across tasks and ratios.

E.4 MMLU

First, the questions from each subject (History, Law, ML) are shuffled by a seed (42). We use the test split of the MMLU dataset to avoid prior exposure of the train set questions to influence the model’s behavior. Next, the questions for each subject are partitioned into pools of up to 150 in-context questions and 20 test questions. Since ML has 112 questions in the MMLU dataset, we save 20 questions for the test pool and use the remaining 92 questions as the in-context pool. This guarantees the in-context and test pools are disjoint.

For each ratio r of task combinations, $n = 30$ samples are independently constructed, drawing 3 questions from the in-context pool according to r and one test set question. We use fewer samples here than for the arithmetic and entity groups for two concrete reasons. First, the MMLU subjects have

small question pools—ML contains only 112 questions, of which 92 remain for the in-context pool after reserving 20 for testing—so keeping n small bounds how often the same questions are reused across the independently constructed samples: at $n = 30$, the pure-MMLU ratio draws at most $3 \times 30 = 90$ in-context questions, within the 92-question ML pool, whereas $n = 50$ would force each question to be reused roughly $1.6\times$ on average, inflating correlation between samples. Second, MMLU prompts are full four-option questions with long stems, making each activation-extraction pass substantially more memory- and compute-intensive than the short arithmetic and entity prompts. We find $n = 30$ sufficient for statistically significant results, as confirmed by the exact permutation null on κ (Appendix B.1).

F Layer sweep projection figures

In the layer-sweep projections below, each figure shows one model, with a panel for each of the eight swept layers (titled by absolute layer index and relative depth). Large outlined dots are the pure-task centroids and small dots the individual samples; the grey $\times$ is the ratio-predicted mixed centroid and the outlined grey dot the actual mixed centroid, joined by a dotted line. Per-ratio κ values are reported in Appendix B.2 rather than on the plots. Colors identify the pure tasks as given in each figure’s legend.

We include two representative layer sweeps—Arithmetic and Entity for Llama 3.2 3B, the model used for the residual-ablation analysis; an MMLU example is shown in Figure 1. The complete set of projections for all five models and all three task groups is available in the accompanying code repository.

Arithmetic — Llama 3.2 3B: LDA projections of contrast vectors by layer

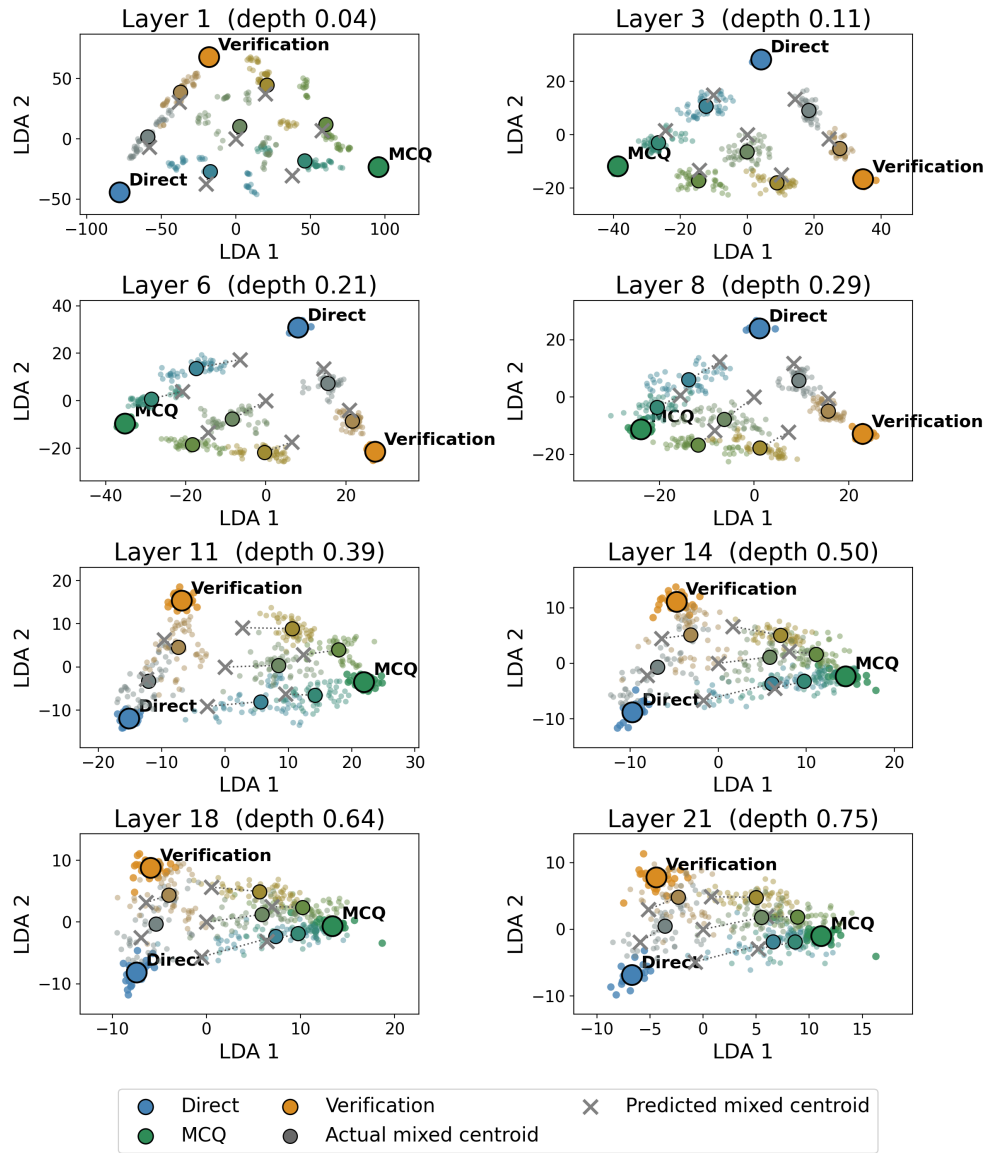


Figure 5: Arithmetic layer sweep projections — Llama 3.2 3B.

Entity — Llama 3.2 3B: LDA projections of contrast vectors by layer

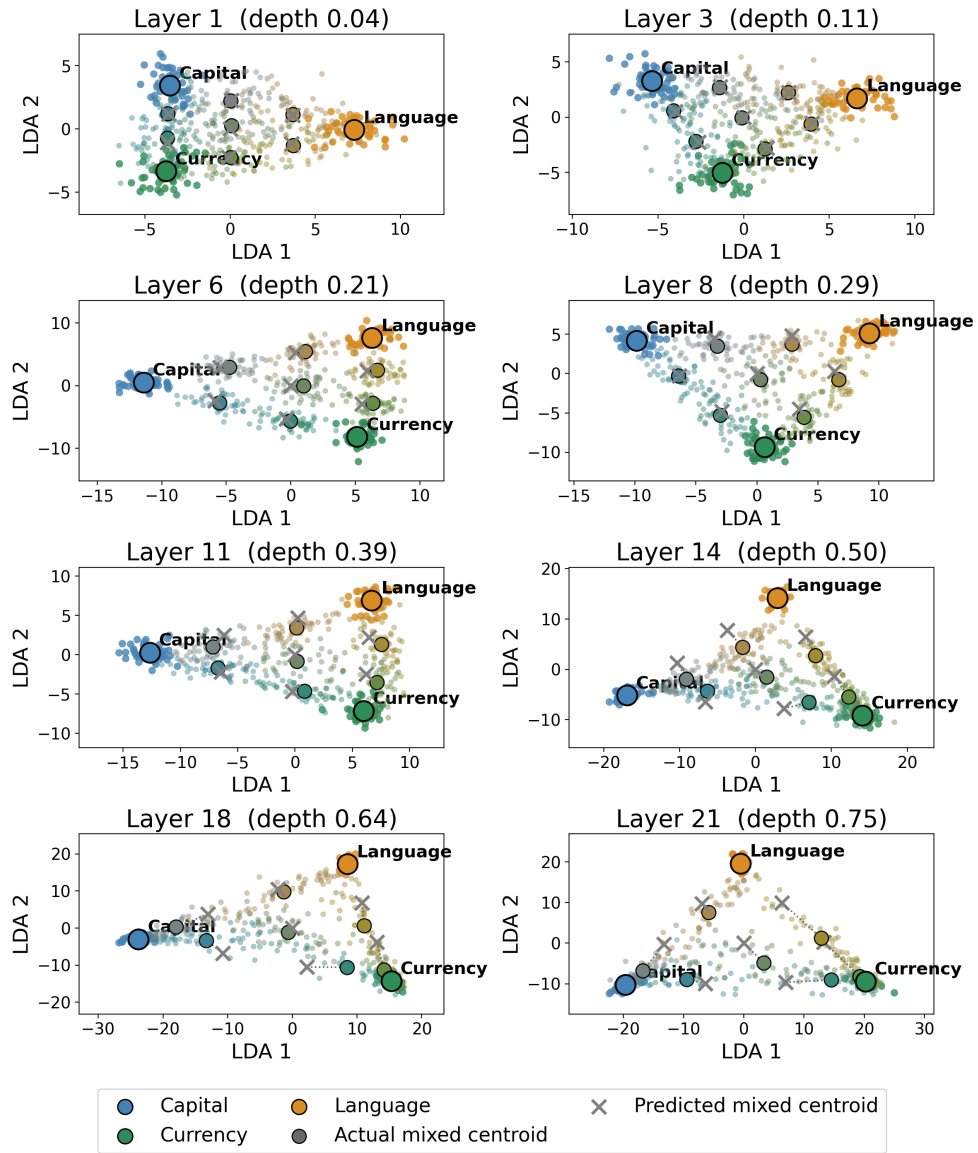


Figure 6: Entity layer sweep projections — Llama 3.2 3B.